

Q1: 24 Feb (Shift 1) - Numerical

4.5 g of compound A (MW = 90) was used to make 250 mL of its aqueous solution. The molarity of the solution in M is $x \times 10^{-1}$. The value of x is _____. (Rounded off to the nearest integer)

Q2: 24 Feb (Shift 2) - Single Correct

Match List-I with List-II

List-I

(a) Valium

(b) Morphine

(c) Norethindrone

(d) Vitamin B₁₂

List-II

(i) Antifertility drug

(ii) Pernicious anaemia

(iii) Analgesic

(iv) Tranquillizer

(1) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

(2) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)

(3) (a) – (ii), (b) – (iv), (c) – (iii), (d) – (i)

(4) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

Q3: 24 Feb (Shift 2) - Numerical

1.86 g of aniline completely reacts to form acetanilide. 10% of the product is lost during purification. Amount of acetanilide obtained after purification (in g) is $\times 10^{-2}$.

Q4: 25 Feb (Shift 1) - Single Correct

Complete combustion of 1.80 g of an oxygen containing compound ($C_xH_yO_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O . The percentage of oxygen in the organic compound is:

(1) 63.53

(2) 53.33

(3) 51.63

(4) 50.33

Q5: 25 Feb (Shift 1) - Numerical

In basic medium CrO_4^{2-} oxidizes $\text{S}_2\text{O}_3^{2-}$ to form SO_4^{2-} and itself changes into $\text{Cr}(\text{OH})_4^-$. The volume of 0.154M CrO_4^{2-} required to react with 40 mL of $0.25\text{M S}_2\text{O}_3^{2-}$ is _____

(Rounded-off to the nearest integer)

Q6: 26 Feb (Shift 1) - Numerical

The number of significant figures in 50000.020×10^{-3} is _____

Q7: 26 Feb (Shift 2) - Numerical

The NaNO_3 weighed out to make 50 mL of an aqueous solution containing 70.0mg Na^+ per mL is _____g.

(Rounded off to the nearest integer) [Given: Atomic weight in g mol^{-1} . Na: 23; N : 14; O : 16]

Answer Key**Q1 (2)****Q2 (4)****Q3 (243)****Q4 (2)****Q5 (173)****Q6 (7)****Q7 (13)**

Q1 (2)

$$\text{Moles of A} = \frac{\text{Weight}}{\text{M.w}}$$

$$= \frac{4.5}{90} = \frac{1}{20} = 0.05$$

$$\text{Volume (Lit)} = \frac{250}{1000} = 0.250 \text{ lit}$$

$$\text{Moles of A} = \frac{\text{Weight}}{\text{M.w}}$$

$$= \frac{4.5}{90} = \frac{1}{20} = 0.05$$

$$\text{Volume (Lit)} = \frac{250}{1000} = 0.250 \text{ lit}$$

$$\text{Molarity (M)} = \frac{\text{Mole}}{(\text{Lit}) \text{ volume}} = \frac{0.05}{0.250} = 0.2$$

$$= 2 \times 10^{-1} \frac{\text{mol}}{\text{Lit}}$$

$$x = 2$$

$$\text{Molarity (M)} = \frac{\text{Mole}}{(\text{Lit}) \text{ volume}} = \frac{0.05}{0.250} = 0.2$$

$$= 2 \times 10^{-1} \frac{\text{mol}}{\text{Lit}} \quad x = 2$$

Q2 (4)

(a) Valium

(iv) Tranquilizer

(b) Morphine

(iii) Analgesic

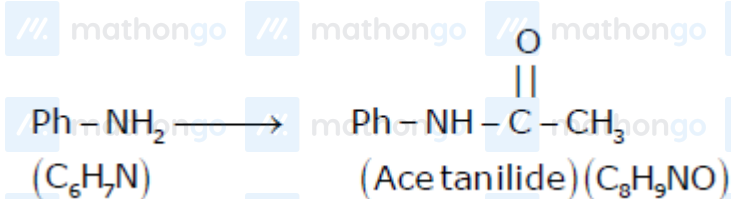
(c) Norethindrone

(i) Antifertility drug

(d) Vitamin B₁₂

(ii) Pernicious anaemia

Q3 (243)



$$\text{Molar mass} = 93 \quad \text{Molar mass} = 135$$

93 g Aniline produce 135 g acetanilide

$$1.86 \text{ g produce } \frac{135 \times 1.86}{93} = 2.70 \text{ g}$$

Hints and Solutions

MathonGo

At 10% loss, 90% product will be formed after purification.

$$\text{Amount of product obtained} = \frac{2.70 \times 90}{100} = 2.43 \text{ g} = 243 \times 10^{-2} \text{ g}$$

Q4 (2)

$$n_{\text{CO}_2} = \frac{2.64}{44} = 0.06$$

$$n_c = 0.06$$

$$\text{weight of carbon} = 0.06 \times 12 = 0.72 \text{ gm}$$

$$n_{\text{H}_2\text{O}} = \frac{1.08}{18} = 0.06$$

$$n_H = 0.06 \times 2 = 0.12$$

$$\text{weight of H} = 0.12 \text{ gm}$$

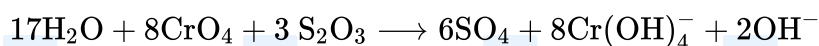
$\therefore$ Weight of oxygen in $\text{C}_x\text{H}_y\text{O}_2$

$$= 1.8 - (0.72 + 0.12)$$

$$= 0.96 \text{ gram \% weight of oxygen} = \frac{0.96}{1.8} \times 100$$

$$= 53.3\%$$

Q5 (173)



$$\text{Applying mole-mole analysis } \frac{0.154 \times V}{8} = \frac{40 \times 0.25}{3}$$

$$V = 173 \text{ mL}$$

Q6 (7)

$$50000.020 \times 10^{-3}$$

Number of significant figure = 7

Q7 (13)

$$\text{Na}^+ = 70 \text{ mg/mL}$$

$$\begin{aligned} W_{\text{Nat}} \text{ in } 50 \text{ mL solution} &= 70 \times 50 \text{ mg} \\ &= 3500 \text{ mg} \end{aligned}$$

$$= 3.5 \text{ gm}$$

$$\text{Moles of Na}^+ \text{ in } 50 \text{ mL solution} = \frac{3.5}{23}$$

Hints and Solutions

MathonGo

Moles of $\text{NaNO}_3 = \text{moles of Na}^+$

$$= \frac{3.5}{23} \text{ mol}$$

$$\text{Mass of NaNO}_3 = \frac{3.5}{23} \times 85 = 12.934$$

$\simeq 13\text{gm Ans.}$